$$e^{ik(x+L)}\, u_k(x+L) = e^{ikx}\, u_k(x)$$

Using Eq. (6.5), we obtain

$$e^{ik(x+L)} = e^{ikx}$$

or $$e^{ikL} = 1$$

which gives

$$k = \frac{2\pi n}{L} \tag{6.45}$$

with $n = \pm 1, \pm 2, \ldots$, or

$$k = \pm \frac{2\pi}{L}, \pm \frac{4\pi}{L}, \ldots \tag{6.46}$$

Therefore, the number of possible wave functions in the range dk is

$$dn = \frac{L}{2\pi} dk \tag{6.47}$$

Considering now the first Brillouin zone, the maximum value of k is π/a where a is the length of the primitive cell. If N is the number of primitive cells in the crystal of length L, then $a = L/N$. Therefore, the maximum value of k in the first Brillouin zone is $\frac{N\pi}{L}$. Thus the series (6.46) is cut off at N/L which means that the total number of allowed k-values in the first Brillouin zone is equal to N. Hence the total number of possible states in any energy band is equal to the number of primitive unit cells N. We further note that since each band can be occupied by a maximum of two electrons, each band can accommodate a maximum of $2N$ electrons. This result is quite important for explaining the distinction between metals, insulators and semiconductors.

6.4 VELOCITY AND EFFECTIVE MASS OF ELECTRON

6.4.1 Velocity of Electron

We consider the velocity of an electron described by a wave vector **k**. According to the wave mechanical theory of particles, a particle moving with a velocity v is equivalent to a wave packet moving with a group velocity v_g. Thus, we have

$$v = v_g = \frac{d\omega}{dk} \tag{6.48}$$

where ω is the angular frequency of the deBroglie waves. Also, the energy E of the particle is given by

$$E = \hbar\omega$$

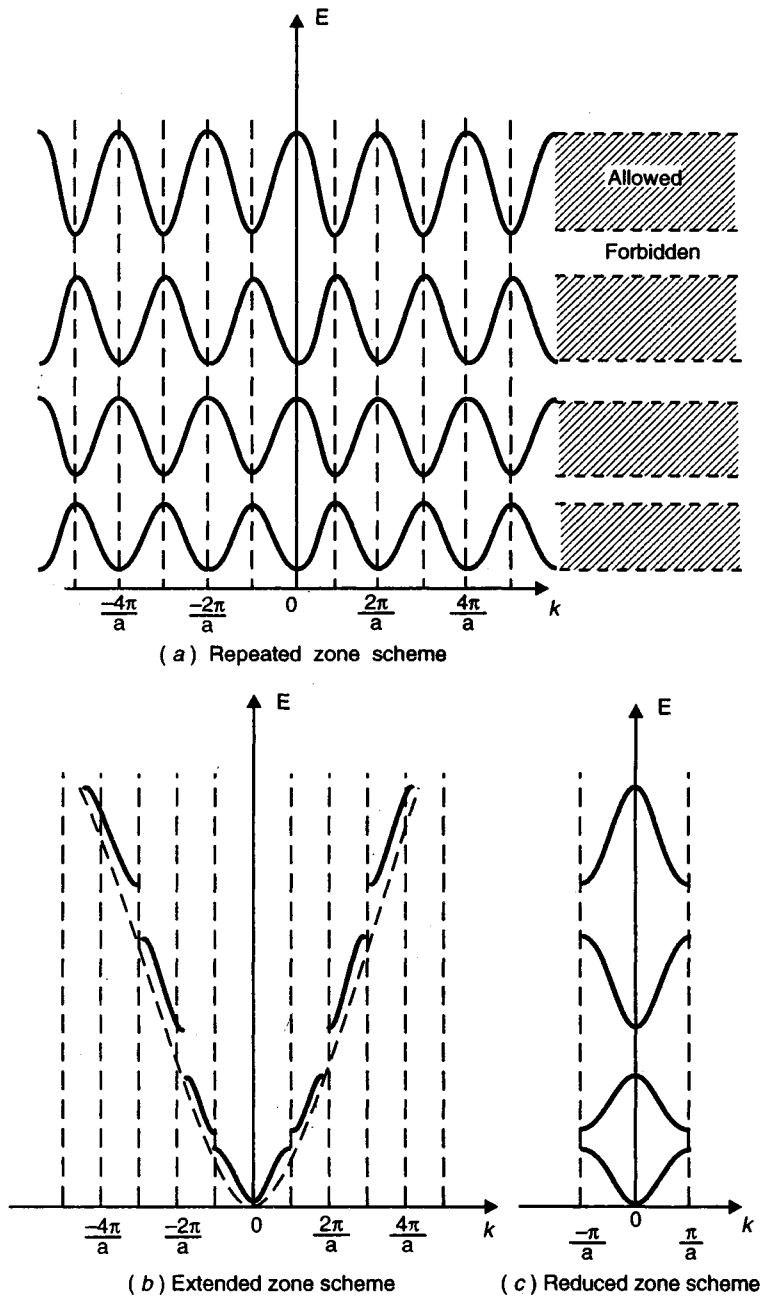


Fig. 6.5. *E versus k curves in three different zone schemes drawn on the same scale. Figure (b) also shows allowed and forbidden energy bands. The dotted parabolic curve in Fig. (b) corresponds to free electrons in a constant potential.*

Therefore, Eq. (6.48) can be written as

$$v = \frac{1}{\hbar}\frac{dE}{dk} \tag{6.49}$$

For free electrons, we have

$$E = \frac{\hbar^2 k^2}{2m}$$

and, therefore, Eq. (6.49) gives

$$v = \frac{\hbar k}{m} = \frac{p}{m}$$

This shows that, for a free electron, v is proportional to k. However, in the band theory, E is generally not proportional to k. The variation of E with k based on the band theory is shown in Fig. 6.6a. This shows that the slope dE/dk of the $E(k)$ curve is not constant but changes with k. Using this curve and employing Eq. (6.49), one can obtain v versus k curve as shown in Fig. 6.6b. This curve indicates that the velocity of the electron is zero for k=0 and $\pm\pi/a$ where the slope dE/dk is zero, i.e., at the top and bottom of the energy band (first Brillouin zone). For $k = k_0$, where k_o corresponds to the inflexion point of the $E(k)$ curve, the absolute value of the velocity attains a maximum value equal to the free electron velocity. Beyond the inflexion point, the velocity decreases with increase in energy which is altogether different from the behaviour of free electrons.

6.4.2 Effective Mass of Electron

Consider now the effect of applied electric field $\mathcal{E}$ on the motion of an electron present in the Brillouin zone. We assume only one electron in the Brillouin zone so that the Pauli's exclusion principle is not violated. If c is the velocity of the electron and the field $\mathcal{E}$ acts on it for a time dt, the increase in energy of the electron is given by

$$dE = e\mathcal{E}dx = e\mathcal{E}vdt = \frac{e\mathcal{E}}{\hbar}\left(\frac{dE}{dk}\right)dt \tag{6.50}$$

We can write

$$dE = \frac{dE}{dk}dk$$

$$\therefore \quad \frac{dE}{dk}dk = \frac{e\mathcal{E}}{\hbar}\left(\frac{dE}{dk}\right)dt$$

or

$$\frac{dk}{dt} = \frac{e\mathcal{E}}{\hbar} \tag{6.51}$$

or
$$\hbar \frac{dk}{dt} = \frac{dp}{dt} = F = e\mathcal{E} \tag{6.52}$$

where p represents the crystal momentum. It follows from Eq. (6.52) that the rate of change of crystal momentum is equal to the impressed force $e\mathcal{E}$. It is thus an analogue to Newton's second law for an electron in a periodic lattice.

The acceleration a of the electron is obtained by differentiating (6.49) with respect to t, i.e.,

$$a = \frac{dv}{dt} = \frac{1}{\hbar} \frac{d^2E}{dk^2} \left(\frac{dk}{dt} \right)$$

Using Eq. (6.51), we obtain

$$a = \frac{e\mathcal{E}}{\hbar^2} \frac{d^2E}{dk^2} \tag{6.53}$$

For a free electron of mass m,

$$a = \frac{e\mathcal{E}}{m} \tag{6.54}$$

A comparison of Eq. (6.53) with (6.54) yields an interesting result of the band theory, i.e., an electron moving in a crystal has an effective mass m* given by

$$m^* = \frac{\hbar^2}{\left(\frac{d^2E}{dk^2} \right)} \tag{6.55}$$

This indicates that the effective mass of an electron is not constant but depends on the value of d^2E/dk^2, i.e., on the shape of the $E(k)$ curve. The variation of the effective mass with k is shown in Fig. 6.6c. The effective mass is positive in the lower half of the energy band and negative in the upper half. It becomes infinite at the inflexion points of the $E(k)$ curve. Thus the electron behaves as a positively charged particle in the upper half of the band only.

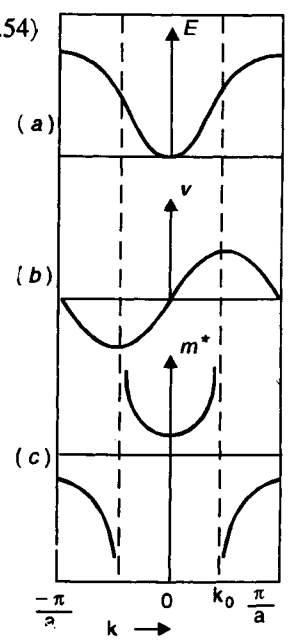


Fig. 6.6. *Variations of energy, velocity and effective mass with k. The dashed lines correspond to inflexion points in the E-k curve.*

It is generally convenient to introduce a factor

$$f_k = \frac{m}{m^*} = \frac{\hbar^2}{\left(\frac{d^2E}{dk^2}\right)} \qquad (6.56)$$

which measures the extent up to which an electron in the k-state behaves as a free electron. If $f_k = 1$, the electron behaves totally as a free electron. If $f_k < 1$, i.e. $m^* > m$, the electron behaves as a heavy particle.

The concept of the negative effective mass may be understood in terms of the Bragg's reflection when k is close to $\pm\pi/a$. Due to the Bragg's reflection, a force applied in one direction leads to a gain of momentum in the opposite direction which results in the negative effective mass.

The concept of effective mass has a physical significance. It provides a satisfactory description of the charge carriers in crystals. In crystals such as alkali metals, which have partially filled energy band, the conduction takes place mainly through electrons. However, in crystals for which the energy band is nearly full, except for a few electron vacancies near the top of the band, these negative charge and negative mass vacancies may be considered as positive charge and positive mass particles called *holes* which act as positive charge carriers to produce conduction.

6.5 DISTINCTION BETWEEN METALS, INSULATORS AND SEMICONDUCTORS

To distinguish between metals, insulators and semiconductors on the basis of the band theory, we consider an energy band which is filled with electrons up to a certain value k_1 ($k_1 < \pi/a$) as shown in Fig. 6.7. It is of interest to know the effective number of free electrons present in this band which gives information about the conductivity of the band. In the expression (6.56), we introduced the term f_k that represents the extent up to which an electron in the k-state behaves as a free electron and hence participates in electrical conduction. Thus, in order to determine the total effective number of free electrons in the energy band, we have to sum up over all the possible f_k's corresponding to the various occupied energy states present in the band. Therefore, we can write

$$N_{eff} = \Sigma f_k \qquad (6.57)$$

The Eq. (6.47) gives the number of possible states in the interval dk for a one-dimensional lattice of length L as

$$dn = \frac{L}{2\pi} dk \qquad (6.47)$$

Since each state is occupied by a maximum of two electrons, the effective number of free electrons present in the band within the limits $-k_1$ and $+k_1$ be-

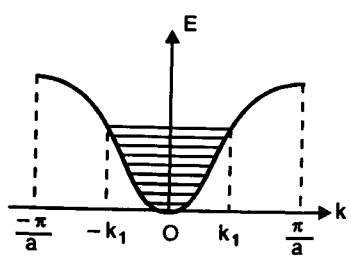


Fig. 6.7. *Energy band (first Brillouin zone) filled up to the state k_1.*

comes

$$N_{eff} = 2\frac{L}{2\pi}\int_{-k_1}^{k_1} f_k dk = \frac{L}{\pi}\int_{-k_1}^{k_1} f_k dk \tag{6.58}$$

Using Eq. (6.56), we obtain

$$N_{eff} = \frac{2L}{\pi}\frac{m}{\hbar^2}\int_0^{k_1}\frac{d^2E}{dk^2}dk$$

$$= \frac{2Lm}{\pi\hbar^2}\left(\frac{dE}{dk}\right)_{k=k_1} \tag{6.59}$$

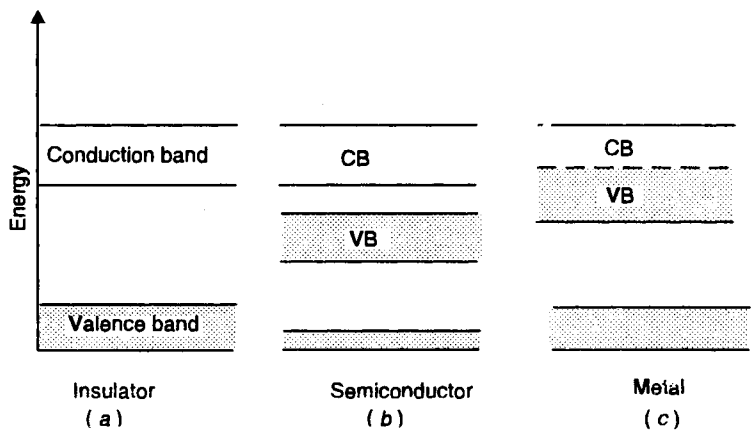


Fig. 6.8. *Electron distribution in an insulator, intrinsic semiconductor and metal at* 0 K. *The shaded regions are occupied by electrons.*

This yields the following results :

(i) The effective number of free electrons in a completely filled band is zero. This is because *dE/dk* vanishes at the top of the band.

(ii) The effective number of electrons attains a maximum value when *dE/dk* becomes maximum, i.e., when the band is filled up to the point of inflexion.

It follows from the above discussion that a solid having a certain number of energy bands completely filled and all other bands completely empty, as shown in Fig. 6.8a, is an insulator. On the other hand, a solid having a partly filled energy band has a metallic character (Fig. 6.8c). The situation depicted in Fig. 6.8a can strictly occur only at 0 K. At temperatures greater than 0 K, some electrons from the topmost filled energy band (valence band)

may get excited to the next higher empty band (conduction band) where they participate in the conduction process. The number of such free or conduction electrons depends on the value of the forbidden energy gap between the valence band and the conduction band. The larger the band gap, the smaller the number of free electrons and hence the larger the tendency of the material to behave as an insulator for all practical purposes. An example of this type of material is diamond which has the band gap of about 6 eV. If the band gap is small ($\cong$1eV) the number of thermally excited electrons becomes appreciable even at ordinary temperatures and the material behaves as an intrinsic semiconductor (Fig. 6.8b). The examples are germanium and silicon. At 0 K, even these materials behave as insulators because electrons from the valence band cannot be thermally excited to the conduction band to cause conduction. Thus it is evident that all the intrinsic semiconductors are insulators at 0 K and all the insulators may behave as semiconductors at temperatures sufficiently higher than 0 K.

It is important to note that the conductivity of semiconductors increases with temperature owing to the increase in the number of conduction band electrons. The conductivity of metals, however, decreases with increase in temperature. This is because more and more phonons are excited at higher temperatures which may scatter conduction electrons and reduce their mobility.

SOLVED EXAMPLES

Example 6.1. Using the Kronig-Penney model, show that for $P << 1$, the energy of the lowest energy band is

$$E = \frac{\hbar^2 P}{ma^2}$$

Solution. Referring to Eq. (6.39), the energy of the lowest band corresponds to $k = \pm \pi/a$, i.e., when

$$P\left(\frac{\sin\alpha a}{\alpha a}\right) + \cos\alpha a = \pm 1$$

Considering only the magnitude on the right hand side, we obtain

$$\frac{P}{\alpha a}(\sin\alpha a) = 1 - \cos\alpha a$$

or

$$\frac{2P}{\alpha a}\sin\left(\frac{\alpha a}{2}\right)\cos\left(\frac{\alpha a}{2}\right) = 2\sin^2\left(\frac{\alpha a}{2}\right)$$

For $P << 1$, we can write

$$\tan\left(\frac{\alpha a}{2}\right) = \frac{P}{\alpha a} = \tan\left(\frac{P}{\alpha a}\right)$$

or $$\frac{\alpha a}{2} = \frac{P}{\alpha a}$$

or $$\alpha^2 = \frac{2P}{a^2}$$

Using Eq. (6.23), we get

$$\frac{2mE}{\hbar^2} = \frac{2P}{a^2}$$

or $$E = \frac{\hbar^2 P}{ma^2}$$

Example 6.2. The energy near the valence band edge of a crystal is given by

$$E = -Ak^2$$

where $A = 10^{-39}$ Jm^2. An electron with wave vector $\mathbf{k} = 10^{10}\,\mathbf{k}_x$ m^{-1} is removed from an orbital in the completely filled valence band. Determine the effective mass, velocity, momentum and energy of the hole.

Solution. We have

$$E = -Ak^2$$

$\therefore$ $$\frac{dE}{dk} = -2Ak,$$

and $$\frac{d^2E}{dk^2} = -2A$$

The effective mass of the electron is

$$m_e^* = \frac{\hbar^2}{\left(\frac{d^2E}{dk^2}\right)}$$

$$= -\frac{\hbar^2}{2A} = -\frac{\left(1.05\times10^{-34}\right)^2}{2\times10^{-39}} = -5.5\times10^{-30}\ \text{kg}.$$

Since the effective mass of a hole is opposite to that of an electron at the same location in the energy band, the effective mass of hole is

$$m_h^* = -m_e^* = 5.5\times10^{-30}\,\text{kg}$$

The wave vector of hole is opposite to that of electron, i.e.,

$$\mathbf{k}_h = -\mathbf{k}_e = -10^{10}\,\hat{\mathbf{k}}_x\ \text{m}^{-1}$$

The momentum of hole is

$$\mathbf{p}_h = \hbar\mathbf{k}_h$$

$$= -1.05\times10^{-34}\times10^{10}\,\hat{\mathbf{k}}_x = -1.05\times10^{-24}\,\hat{\mathbf{k}}_x\ \text{Jsm}^{-1}$$

The velocity of hole is

$$v_h = \frac{\mathbf{p}_h}{m_h^*} = -\frac{1.05\times10^{-24}\,\hat{\mathbf{k}}_x}{5.5\times10^{-30}} = -1.9\times10^5\,\hat{\mathbf{k}}_x\ \text{ms}^{-1}$$

The energy of the electron with wave vector $\mathbf{k}_e$ is

$$E_e = -Ak^2{}_e$$

$$= -10^{-39}(10^{10}\,\hat{\mathbf{k}}_x)^2$$

$$= -10^{-19}\ \text{J}$$

Therefore, the energy of the hole referred to zero at the top of the valence band is

$$E_h = -E_e$$

$$= 10^{-19}\ \text{J}$$

SUMMARY

1. The band theory assumes that the potential experienced by an electron in a crystal is periodic with period equal to the lattice constant.

2. According to the Bloch theorem, the solution to the Schrodinger wave equation for an electron moving in a periodic potential is of the form of the Bloch function, i.e.,

$$\psi_k(\mathbf{r}) = e^{i\mathbf{k}.\mathbf{r}}\,u_k(\mathbf{r})$$

where $u_k(\mathbf{r})$ is invariant under a crystal lattice translation.

3. The energy spectrum of an electron moving in a periodic potential consists of a set of allowed energy regions or bands which are separated by forbidden energy regions called the band gaps. The forbidden energy regions are those in which **k** assumes complex values and no Bloch function type solutions to the wave equation are possible.

4. The width of the allowed energy bands increases with increase in total energy and decrease in binding energy of the electrons. For very large binding energies, the energy bands are transformed into energy levels. If the electrons are completely free, the energy spectrum is quasi-continuous.

5. The *E-k* curve in the extended zone scheme has discontinuities at

$$k = \pm\frac{n\pi}{a}$$

where a is the length of the primitive cell and $n = 1, 2, 3, \ldots$ These k-values define the boundaries of the Brillouin zones which produce Bragg reflection of electrons.

6. In a crystal having N primitive cells, the maximum number of energy states per band is N and the maximum number of electrons per band is $2N$.

7. The velocity of an electron is given by the expression

$$v = \frac{1}{\hbar}\frac{dE}{dk}$$

It becomes maximum at the inflextion points of the *E–k* curve where it approaches the free electron velocity. Beyond the inflexion points, the velocity decreases with increase in energy.

8. The effective mass of an electron moving in a crystal depends on the shape of the *E–k* curve and is given by the expression

$$m^* = \frac{\hbar^2}{\left(\frac{d^2E}{dk^2}\right)}$$

In the first Brillouin zone, the effective mass is positive in the lower half and negative in the upper half of the energy band. It beeomes infinite at the inflexion points of the *E–k* curve.

9. For a one-dimensional lattice of length L, the total effective number of free electrons in an energy band filled with electrons up to a certain value k_1 ($k_1 < \pi/a$) is

$$N_{eff} = \frac{2Lm}{\pi\hbar^2}\left(\frac{dE}{dk}\right)_{k=k_1}$$

where m is the mass of an electron. N_{eff} is zero at the top of the band and attains a maximum value at the points of inflexion.

VERY SHORT QUESTIONS

1. What is the Bloch function?
2. What is forbidden energy gap?
3. Can an electron possess negative effective mass? Justify.
4. Give the order of band gap for a metal, a semiconductor and an insulator.
5. Why a solid whose energy bands is filled can not be a metal?

SHORT QUESTIONS

1. What are the shortcomings of the free electron theory? What is the main cause of its failure?
2. Compare the salient features of the band theory vis-a-vis the free electron theory. Discuss the successes and failures of these theories.
3. What is the Bloch theorem ?
4. Explain the concept of forbidden energy bands.
5. Distinguish between a metal, a semiconductor and an insulator on the basis of their energy band structure.
6. Show that the existence of energy gaps at the boundaries of the Brillouin zone of a one-dimensional lattice is equivalent to the condition of Bragg reflection of electron waves.
7. Show the dependence of velocity of electron on wave vector **k** as predicted by the band theory.
8. Prove that the number of different **k**-states in the first Brillouin zone of a simple cubic lattice is equal to the number of lattice sites.
9. Prove that the total number of possible states in an energy band of a finite crystal is equal to the number of primitive cells in it.
10. How does the band theory lead to the concept of negative effective mass ?
11. The conductivity of metals decreases while that of semiconductors increases with rise in temperature. Explain.

LONG QUESTIONS

1. State and prove the Bloch theorem. Discuss its importance in the band theory.
2. Discuss the formation of allowed and forbidden energy bands on the basis of the Kronig-Penney model. Discuss the extreme conditions when energy levels are either discrete or continuous. What is the effect of changing the binding energy of electron on the energy bands ?
3. Prove that the motion of electrons through the periodic potential in solids gives rise to the band structure.
4. What are Bloch functions? Explain the origin of allowed and forbidden bands for electrons in solids. What is the number of orbitals in an energy band ?
5. Describe the periodic zone scheme, extended zone scheme and reduced zone scheme for representing E-k relationships.
6. Discuss the motion of electrons in one-dimension according to the band theory and show the variation of energy, velocity and effective mass as a function of wave-vector.
7. What is meant by the effective mass of an electron? What is its significance? Show that the effective mass of an electron in a crystal is inversely proportional to the second derivative of the E-k curve. Discuss the conditions when the effective mass of an electron becomes positive, negative and infinity.

PROBLEMS

1. The potential of an electron in a one-dimensional lattice is of the same type as that used in the Kronig-Penney model. Assuming

$$V_0\, ab << \hbar^2/m,$$

prove that the energy band gap at $k = \pi/a$ is $2V_0 b/a$.

2. A one-dimensional lattice of spacing a has a potential distribution of the type as considered in the Kronig-Penney model. The value of the potential is $-V$ at each lattice point and abruptly changes to zero at a distance of $0.1a$ on either side of the lattice point. Determine the width of the first energy gap in the electron energy spectrum.

(0.37 V)